

# Oil Pan



| Material                                                                                                                                                  |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Item                                                                                                                                                      | Specification     |
| Motorcraft® High Performance Engine RTV Silicone<br>TA-357                                                                                                | WSE-<br>M4G323-A6 |
| Motorcraft® Metal Surface Prep Wipes<br>ZC-31-B                                                                                                           | —                 |
| Motorcraft® SAE 5W-30 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada)<br>XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada) | WSS-<br>M2C946-A  |
| Motorcraft® Silicone Gasket Remover<br>ZC-30-A                                                                                                            | —                 |

## Removal

**NOTICE:** During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces, that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

## All vehicles

1. **NOTE:** This step does not apply if the front engine cover is removed.

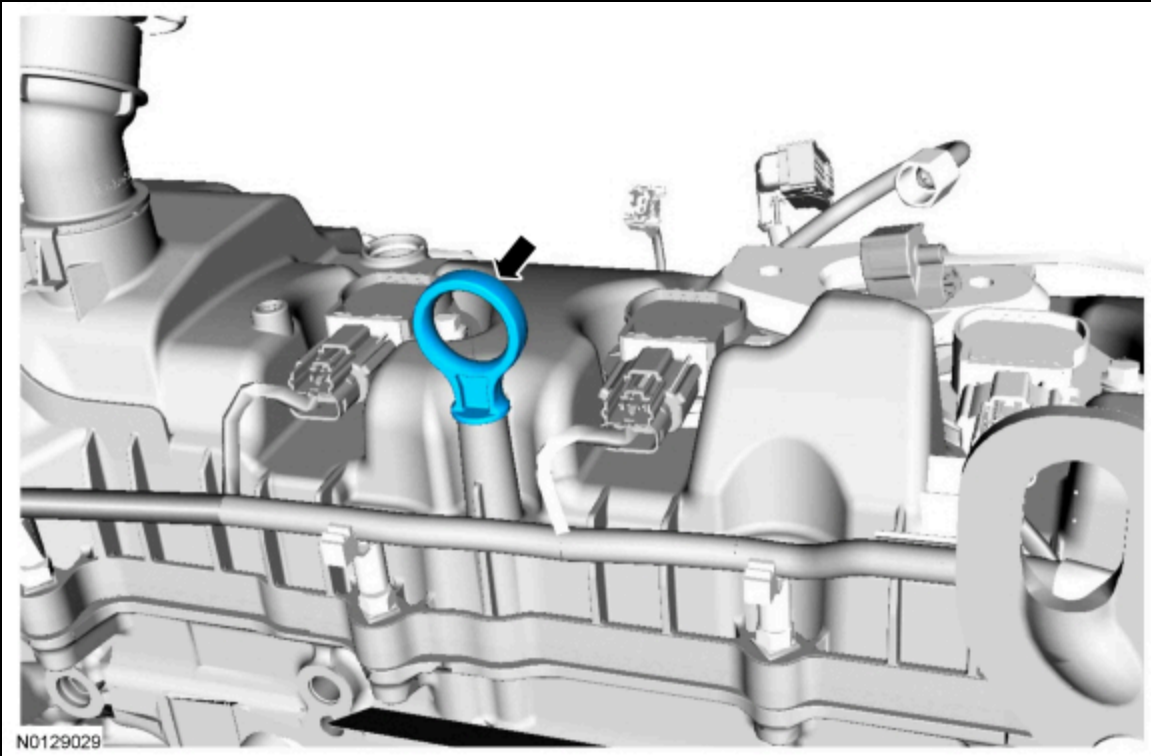
With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.

2. **NOTE:** This step does not apply if the front engine cover is removed.

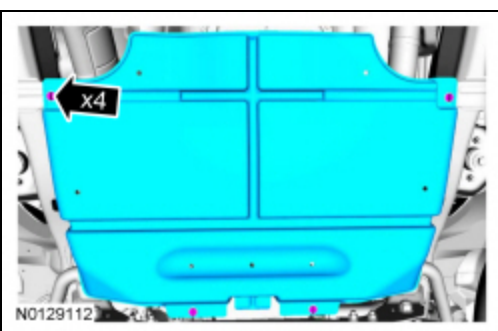
Disconnect the battery ground cable. For additional information, refer to Section 414-01.

3. **NOTE:** This step does not apply if the front engine cover is removed.

Remove the oil level indicator.

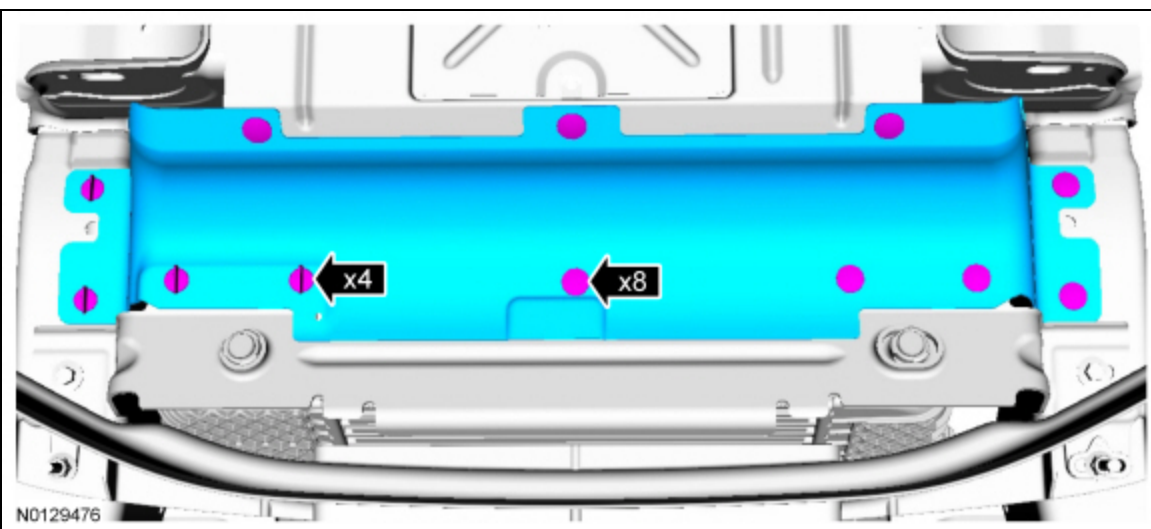


4. Release the 4 fasteners and remove the splash shield.



5. **NOTE:** *This step does not apply if the front engine cover is removed.*

If equipped, turn the 4 quarter-turn retainers, detach the 8 pushpins and remove the lower air deflector.



6. If equipped, Remove the 4 bolts and the skid plate.

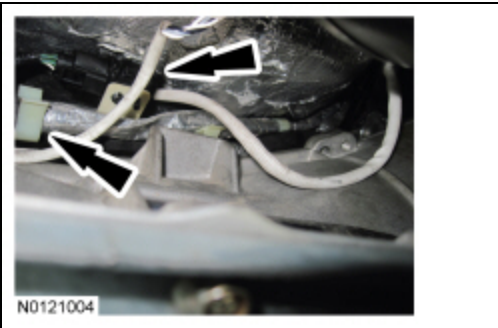
7. Remove the front air dam. For additional information, refer to the Front End Body Panels — Exploded View in Section 501-02.

8. **NOTE:** *This step does not apply if the front engine cover is removed.*

**NOTE:** *Only one wire harness retainer and one electrical connector shown, others similar.*

Disconnect the 2 HO2S electrical connectors.

- Disconnect the 2 HO2S electrical connector retainers.
- If equipped, disconnect the 5 transmission wire harness retainers from the transfer case wire harness.



9. **NOTE:** *This step does not apply if the front engine cover is removed.*

Disconnect the HO2S electrical connector at the LH rear of the transmission.

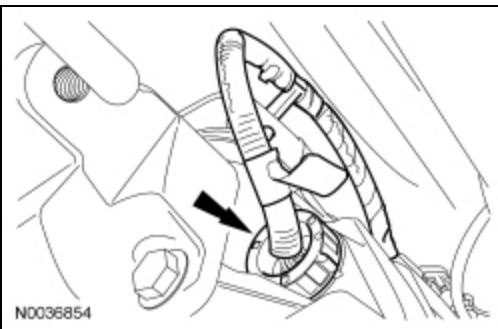
- Disconnect the electrical connector retainer from the bracket at the rear of the transmission.



10. **NOTICE:** **Do not pull on the wire harness to disconnect the connector or damage to the connector will occur.**

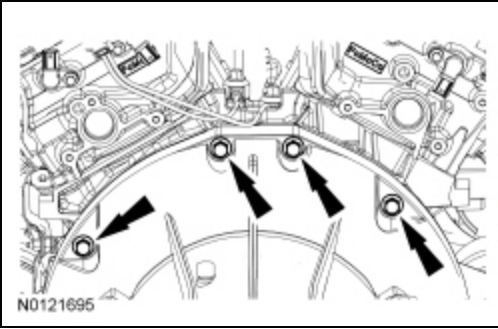
**NOTE:** *This step does not apply if the front engine cover is removed.*

Disconnect the transmission vehicle harness connector by twisting the outer shell and pulling back on the connector.

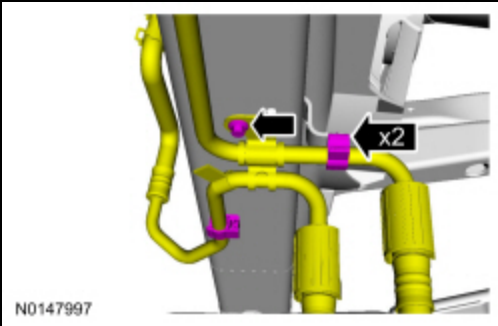


11. **NOTE:** *This step does not apply if the front engine cover is removed.*

Loosen the 4 upper bellhousing-to-engine bolts 5 mm (0.19 in).



12. Remove the bolt and disconnect the 2 retainers for the transmission cooler tubes.



13. **NOTE:** *This step does not apply if the front engine cover is removed.*

Remove the drain plug and drain the engine oil.

- Install the drain plug and tighten to 27 Nm (20 lb-ft).

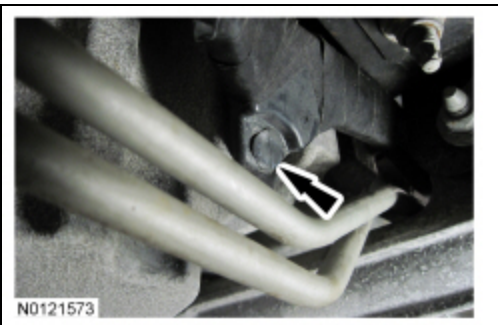


14. Remove and discard the engine oil filter.

15. Remove the starter motor. For additional information, refer to Section 303-06.

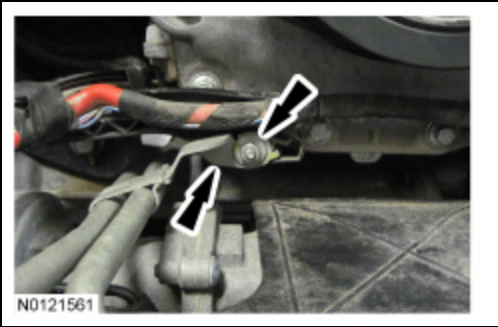
16. **NOTE:** *This step does not apply if the front engine cover is removed.*

Detach the starter motor wiring harness retainer from the oil pan.



17. **NOTE:** *This step does not apply if the front engine cover is removed.*

Remove the nut and the transmission cooler line bracket from the engine front cover stud bolt.

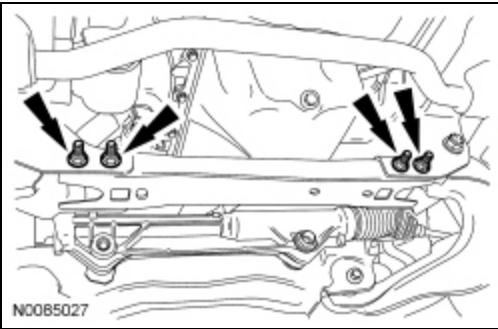


18. **NOTE:** *This step does not apply if the front engine cover is removed.*

Detach the generator wiring harness from the engine front cover.

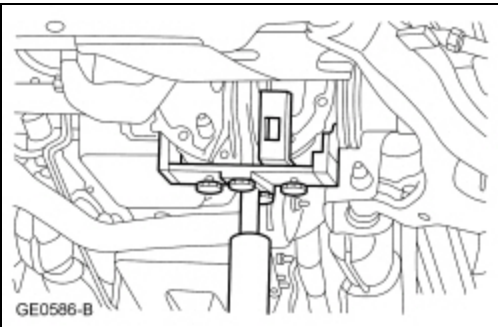


19. Remove the 4 nuts and bolts and the crossmember.

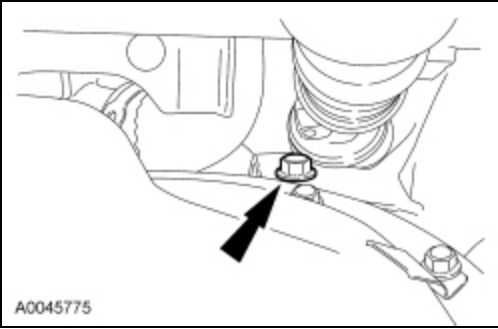


#### **Four Wheel Drive (4X4) vehicles**

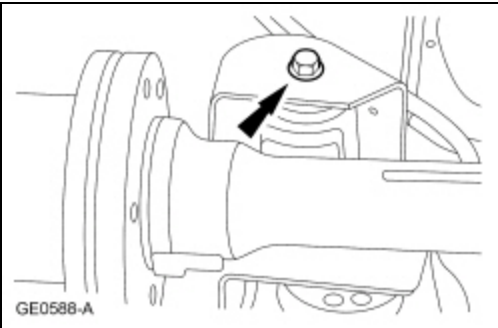
20. Position a suitable hydraulic jack under the front axle. Securely strap the axle to the jack.



21. Remove the upper front axle carrier mounting bushing bolt.

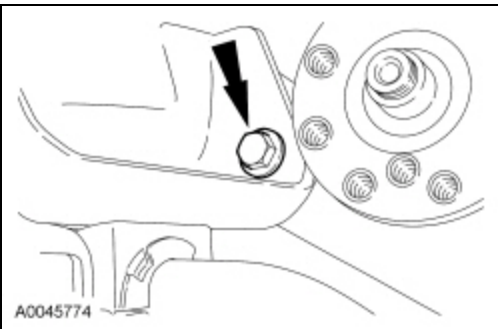


22. Remove the axle shaft housing carrier bushing bolt.



23. **NOTE:** *Front driveshaft removed for clarity only.*

Remove the lower front axle carrier mounting bushing bolt.



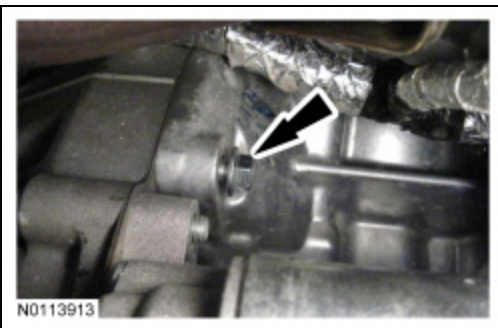
24. **NOTICE:** Use care when lowering the front axle housing, or the vacuum lines to the axle solenoid may become disconnected or damaged.

Lower the axle to allow clearance for the oil pan to be removed.

# **All vehicles**

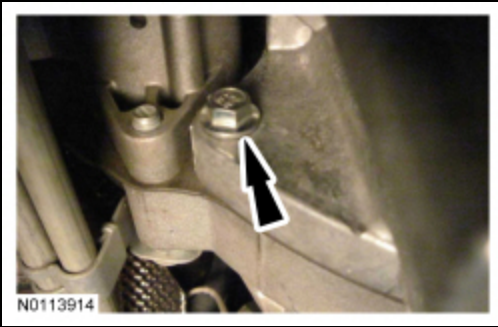
25. **NOTE:** *This step does not apply if the front engine cover is removed.*

Loosen the engine-to-bellhousing bolt above the starter motor 5 mm (0.19 in).



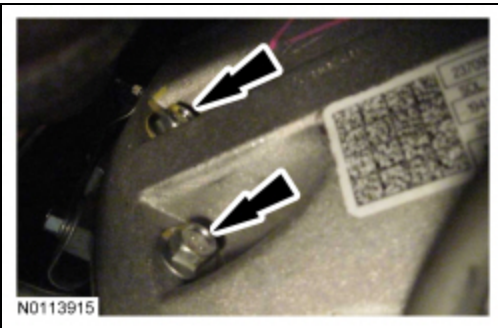
26. Remove the oil pan-to-bellhousing bolt from below the starter motor.



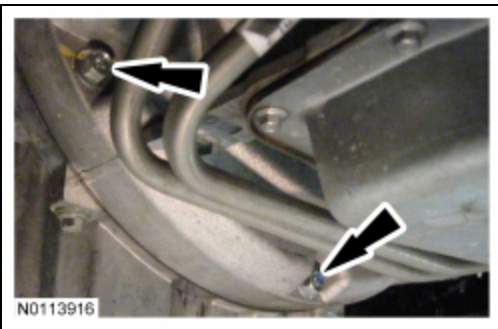


27. **NOTE:** *This step does not apply if the front engine cover is removed.*

Loosen the 2 LH bellhousing-to-engine bolts 5 mm (0.19 in).

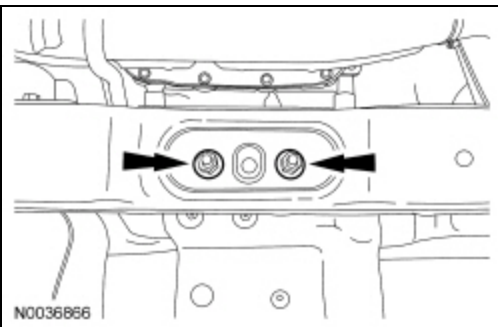


28. Remove the 2 LH bellhousing-to-oil pan bolts.



29. **NOTE:** *This step does not apply if the front engine cover is removed.*

Loosen the 2 transmission crossmember nuts.

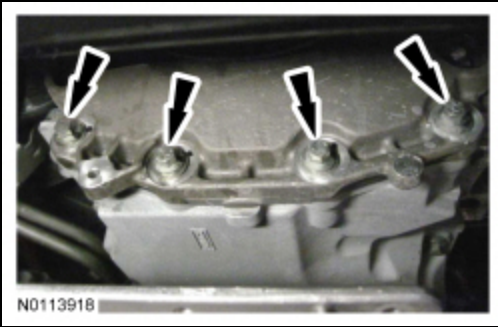


30. **NOTE:** *This step does not apply if the front engine cover is removed.*

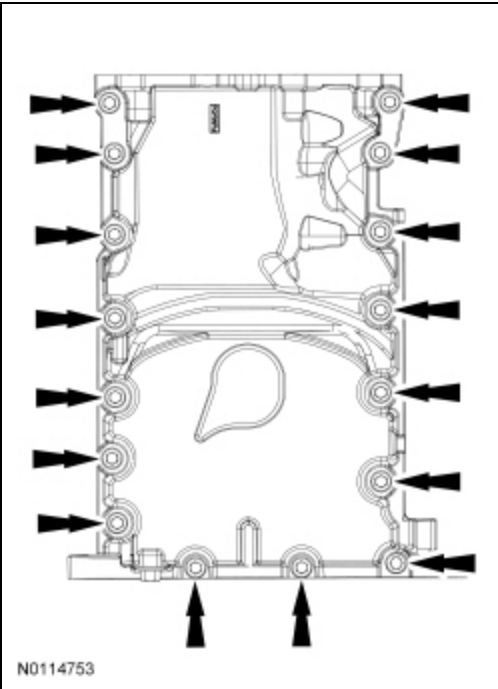
Slide the transmission rearward 5 mm (0.19 in).

31. **NOTE:** *This step does not apply if the front engine cover is removed.*

Remove the 4 lower engine front cover bolts.

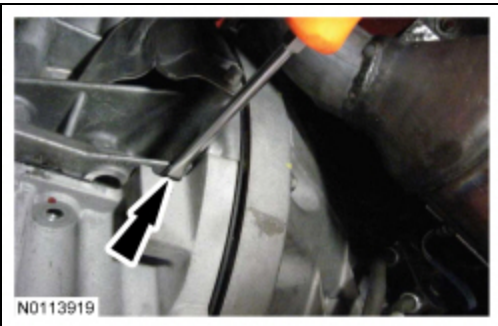


32. Remove the 16 oil pan bolts.



33. **NOTE:** *LH rear shown, RH front similar.*

Using a suitable pry tool, locate the 2 pry pads at the LH and RH sides of the oil pan and pry the oil pan loose and remove.



34. **NOTE:** *The generator wiring harness retainer is being removed for clearance to install the oil pan.*

Detach the generator wiring harness retainer from the engine block.





- 35. NOTICE:** Only use a 3M™ Roloc® Bristle Disk (2-in white, part number 07528) to clean the oil pan. Do not use metal scrapers, wire brushes or any other power abrasive disk to clean. These tools cause scratches and gouges that make leak paths.

Clean the engine oil pan using a 3M™ Roloc® Bristle Disk (2-in white, part number 07528) in a suitable tool turning at the recommended speed of 15,000 rpm.

- Thoroughly wash the oil pan to remove any foreign material, including any abrasive particles created during the cleaning process.

- 36. NOTICE:** Do not use wire brushes, power abrasive discs or 3M™ Roloc® Bristle Disk (2-in white, part number 07528) to clean the cylinder block and engine front cover sealing surfaces. These tools cause scratches and gouges that make leak paths. They also cause contamination that causes premature engine failure. Remove all traces of the gasket.

Clean the sealing surfaces of the cylinder block and engine front cover in the following sequence.

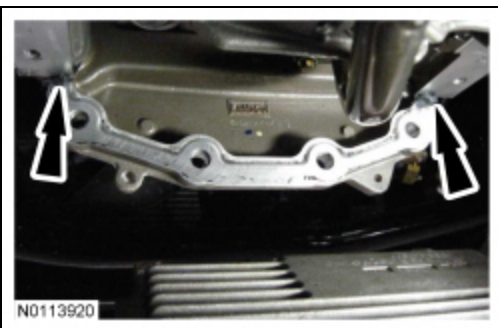
1. Remove any large deposits of silicone or gasket material.
2. Apply silicone gasket remover and allow to set for several minutes.
3. Remove the silicone gasket remover. A second application of silicone gasket remover may be required if residual traces of silicone or gasket material remain.
4. Apply metal surface prep to remove any remaining traces of oil and to prepare the surfaces to bond. Do not attempt to make the metal shiny. Some staining of the metal surfaces is normal.

## Installation

### All vehicles

- 1. NOTE:** This step does not apply if the front engine cover is removed.

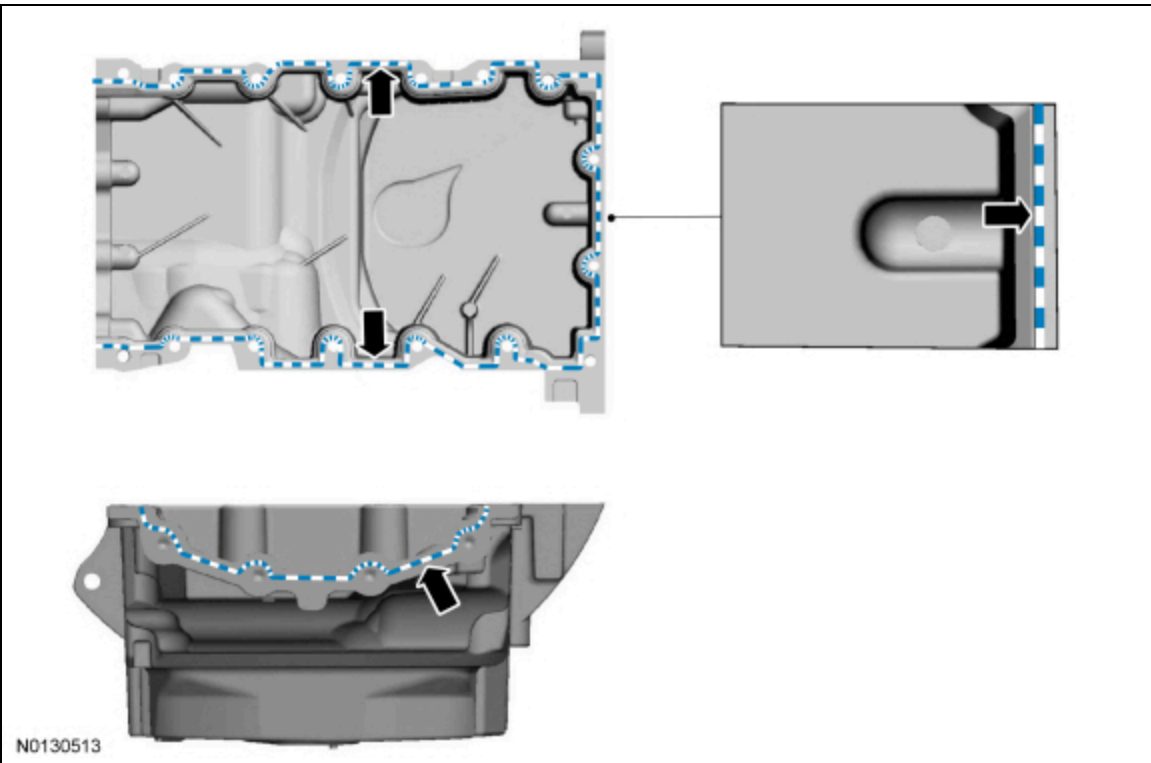
Apply a 5.5 mm (0.21 in) bead of Motorcraft® High Performance Engine RTV Silicone to the 2 engine front cover-to-cylinder block joint areas on the sealing surface of the oil pan.



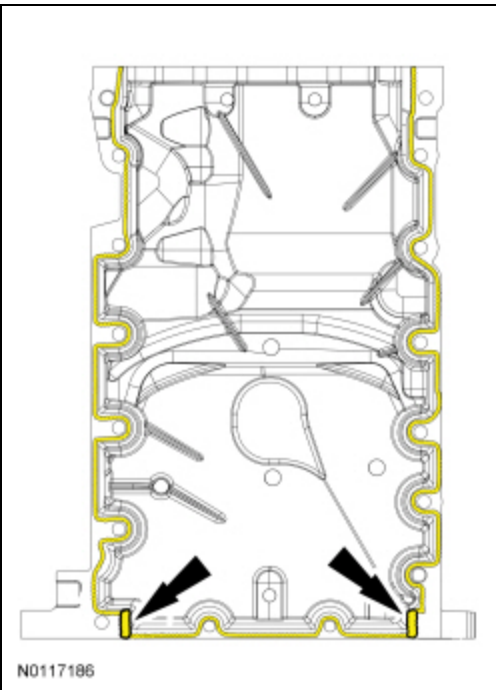
- 2. NOTICE:** Failure to use Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.

**NOTE:** The oil pan and the 4 specified bolts must be installed and the oil pan aligned to the cylinder block within 4 minutes of sealant application. Final tightening of the oil pan bolts must be carried out within 60 minutes of sealant application.

Apply a 3 mm (0.11 in) bead of Motorcraft® High Performance Engine RTV Silicone on the inside edge of the oil pan to the sealing surface of the oil pan-to-engine block mating surface.



3. Apply a 5.5 mm (0.21 in) bead of Motorcraft® High Performance Engine RTV Silicone to the 2 crankshaft seal retainer plate-to-cylinder block joint areas on the sealing surface of the oil pan.

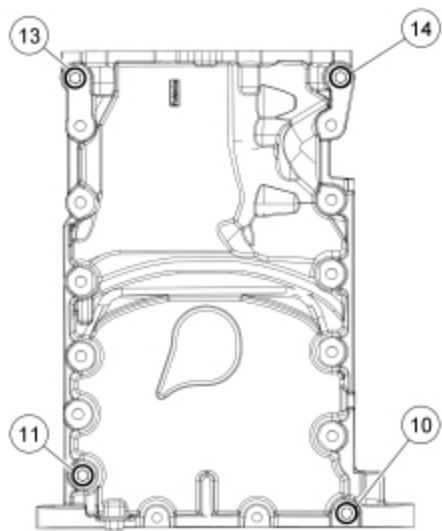


4. **NOTE:** The oil pan and the 4 specified bolts must be installed within 4 minutes of the start of sealant application.

**NOTE:** Keep the oil pan as close as possible to the transmission while installing, then slide forward towards the engine front cover to prevent wiping off of the sealant.

Install the oil pan and bolts 10, 11, 13 and 14.

- Tighten finger-tight.

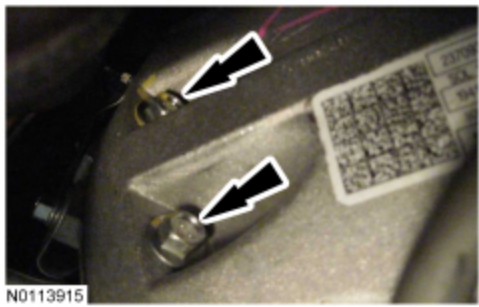


N0114563

5. **NOTE:** *This step does not apply if the front engine cover is removed.*

Install the 2 LH bellhousing-to-engine bolts.

- Do not torque at this time.

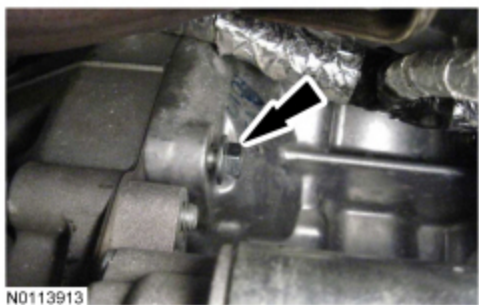


N0113915

6. **NOTE:** *This step does not apply if the front engine cover is removed.*

Install the RH engine-to-bellhousing bolt.

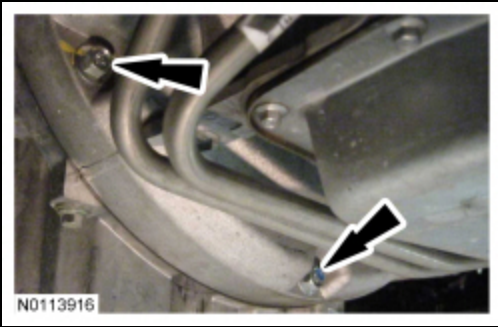
- Do not torque at this time.



N0113913

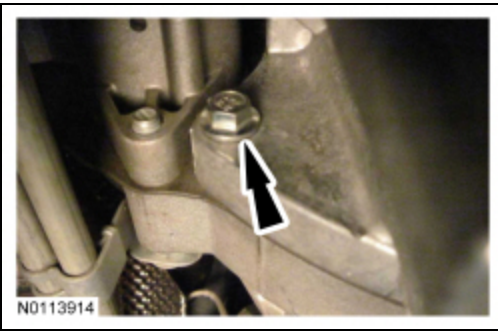
7. Install the 2 LH bellhousing-to-oil pan bolts.

- Do not torque at this time.

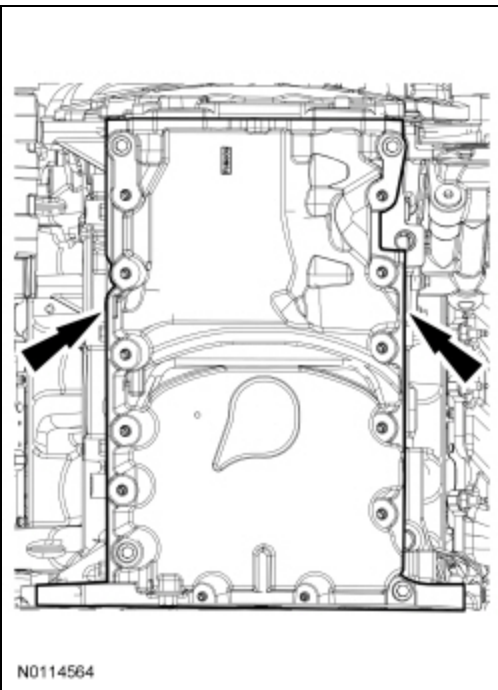


8. Install the RH oil pan-to-bellhousing bolt.

- Do not torque at this time.

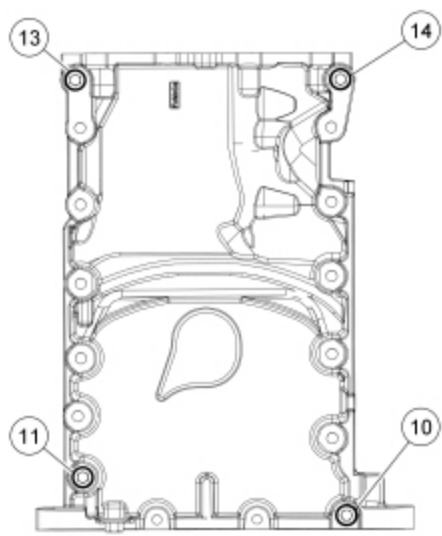


9. Laterally align the oil pan to the block. Make sure the oil pan is flush with the block at these 2 points.



10. Tighten bolts 10, 11, 13 and 14 in the sequence shown.

- Tighten to 3 Nm (27 lb-in).

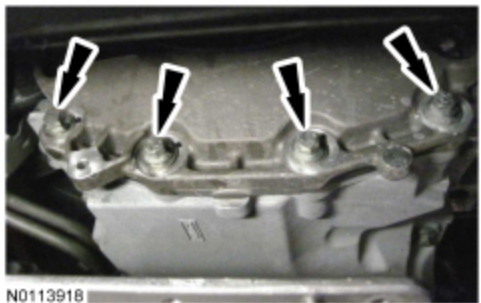


N0114563

11. **NOTE:** *This step does not apply if the front engine cover is removed.*

Install the 4 lower engine front cover bolts.

- Tighten finger-tight.

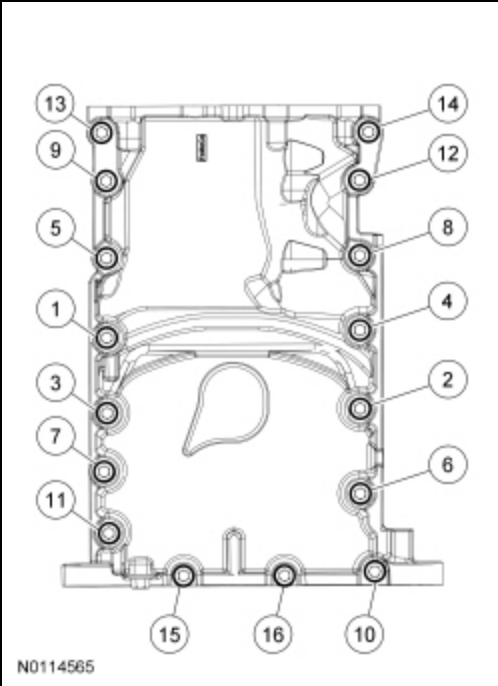


N0113918

12. Install the remaining oil pan bolts. Tighten in 3 stages in the sequence shown.

- Stage 1: Tighten bolts 1-9 and 11-14 to 20 Nm (177 lb-in), then rotate an additional 45 degrees.
- Stage 2: Tighten bolts 15 and 16 to 10 Nm (89 lb-in), then rotate an additional 45 degrees.
- Stage 3: Tighten bolt 10 to 20 Nm (177 lb-in), then rotate an additional 90 degrees.

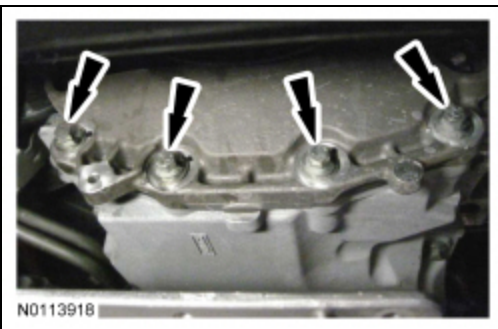




13. **NOTE:** *This step does not apply if the front engine cover is removed.*

Tighten the 4 lower engine front cover bolts.

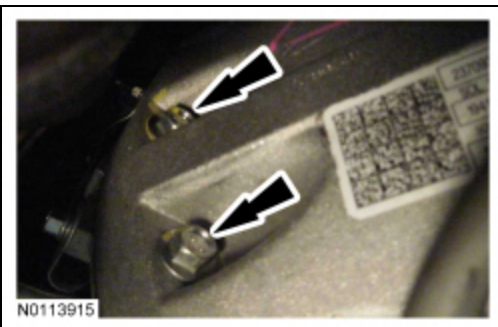
- Tighten to 24 Nm (18 lb-ft).



14. **NOTE:** *This step does not apply if the front engine cover is removed.*

Tighten the 2 LH bellhousing-to-engine bolts.

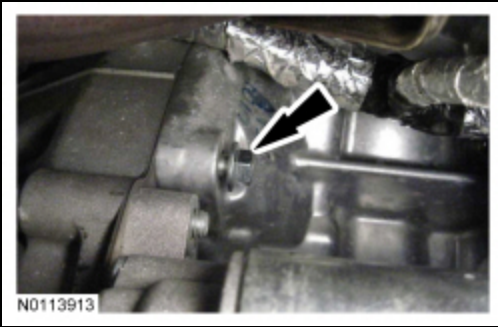
- Tighten to 48 Nm (35 lb-ft).



15. **NOTE:** *This step does not apply if the front engine cover is removed.*

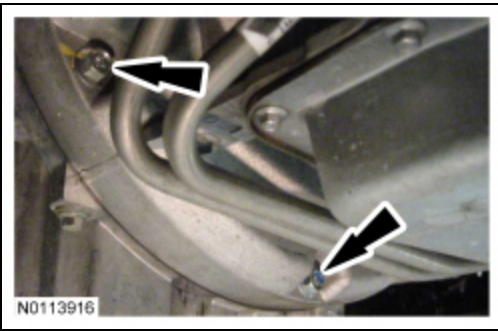
Tighten the RH engine-to-bellhousing bolt.

- Tighten to 48 Nm (35 lb-ft).



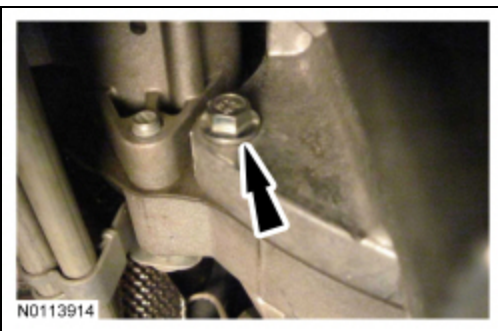
16. Tighten the 2 LH bellhousing-to-oil pan bolts.

- Tighten to 48 Nm (35 lb-ft).



17. Tighten the RH oil pan-to-bellhousing bolt.

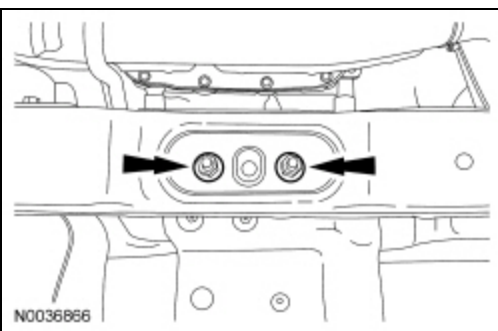
- Tighten to 48 Nm (35 lb-ft).



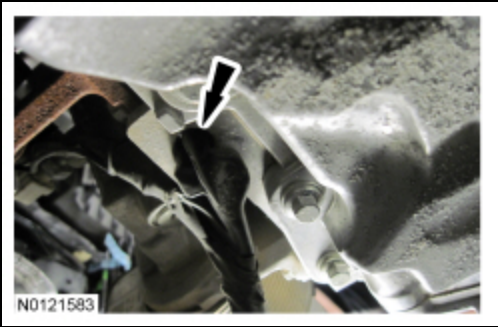
18. **NOTE:** *This step does not apply if the front engine cover is removed.*

Tighten the 2 transmission crossmember nuts.

- Tighten to 103 Nm (75 lb-ft).



19. Attach the generator wiring harness retainer to the engine block.

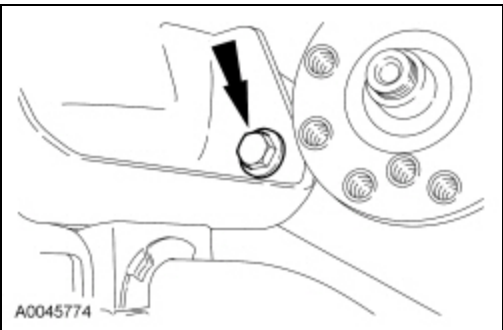


**Four Wheel Drive (4X4) vehicles**

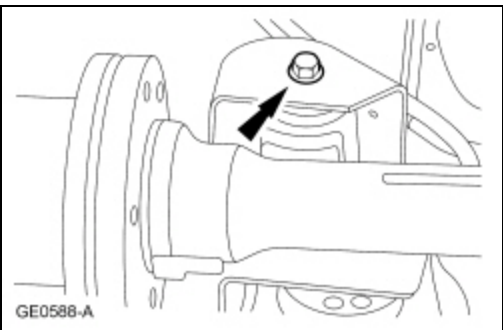
20. **NOTICE:** Use care when positioning the front axle housing, or the vacuum lines to the axle solenoid may become disconnected or damaged.

Raise the front axle carrier into position.

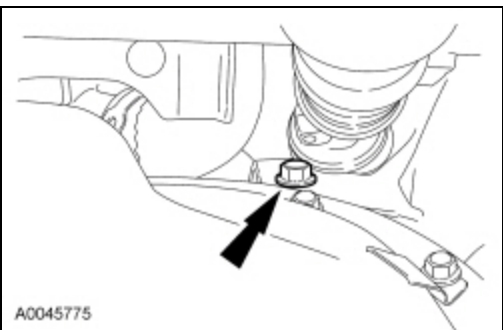
21. Install the lower front axle carrier mounting bushing bolt.
- Tighten to 115 Nm (85 lb-ft).



22. Install the axle shaft housing carrier bushing bolt.
- Tighten to 115 Nm (85 lb-ft).



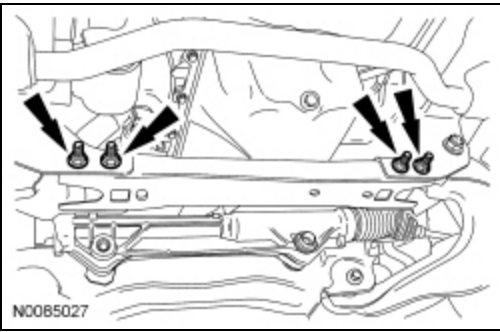
23. Install the upper front axle carrier mounting bushing bolt.
- Tighten to 115 Nm (85 lb-ft).



**All vehicles**

24. Position the crossmember and install the 4 nuts and bolts.

- Tighten to 90 Nm (66 lb-ft).



25. **NOTE:** *This step does not apply if the front engine cover is removed.*

Attach the generator wiring harness to the engine front cover.



26. **NOTE:** *This step does not apply if the front engine cover is removed.*

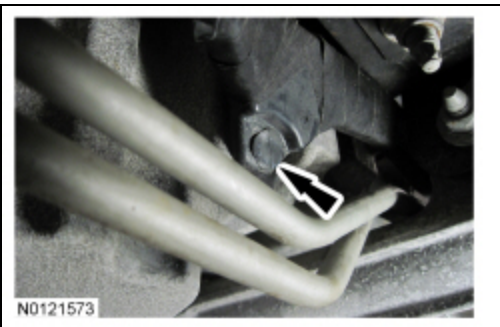
Install the transmission cooler line bracket and nut to the engine front cover stud bolt.

- Tighten to 12 Nm (106 lb-in).



27. **NOTE:** *This step does not apply if the front engine cover is removed.*

Attach the starter motor wiring harness retainer to the oil pan.



28. Install the starter motor. For additional information, refer to Section 303-06.

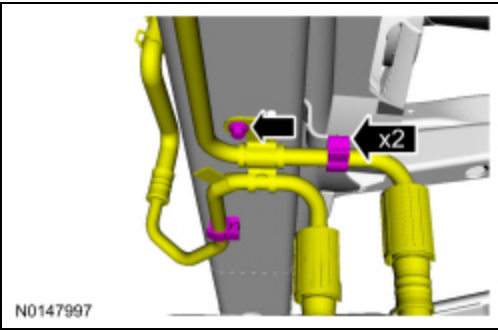
29. **NOTE:** *Do not lubricate the engine oil filter gasket.*

Install a new engine oil filter.

- Tighten to 5 Nm (44 lb-in) and then rotate an additional 180 degrees.

30. Position the transmission cooler tubes and connect the 2 retainers. Install the bolt for the transmission cooler tube bracket.

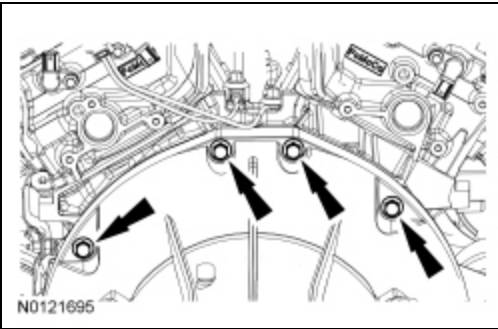
- Tighten to 12 Nm (106 lb-in).



31. **NOTE:** *This step does not apply if the front engine cover is removed.*

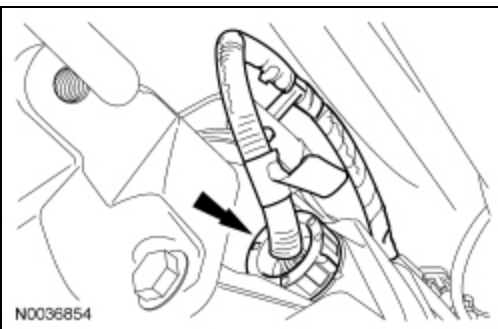
Tighten the 4 upper bellhousing-to-engine bolts.

- Tighten to 48 Nm (35 lb-ft).



32. **NOTE:** *This step does not apply if the front engine cover is removed.*

Connect the transmission vehicle harness connector by pushing the connector into the transmission and twisting the outer shell.



33. **NOTE:** *This step does not apply if the front engine cover is removed.*

Connect the electrical connector retainer to the bracket at the rear of the transmission.

- Connect the Heated Oxygen Sensor (HO2S) electrical connector.



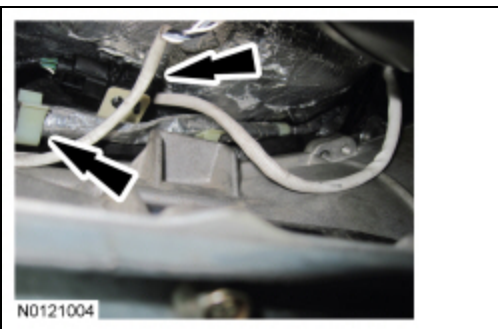


34. **NOTE:** *This step does not apply if the front engine cover is removed.*

**NOTE:** *Only one wire harness retainer and one electrical connector shown, others similar.*

If equipped, connect the 5 transmission wire harness retainers to the transfer case wire harness.

- Connect the 2 HQ2S electrical connector retainers.
- Connect the 2 HQ2S electrical connectors.



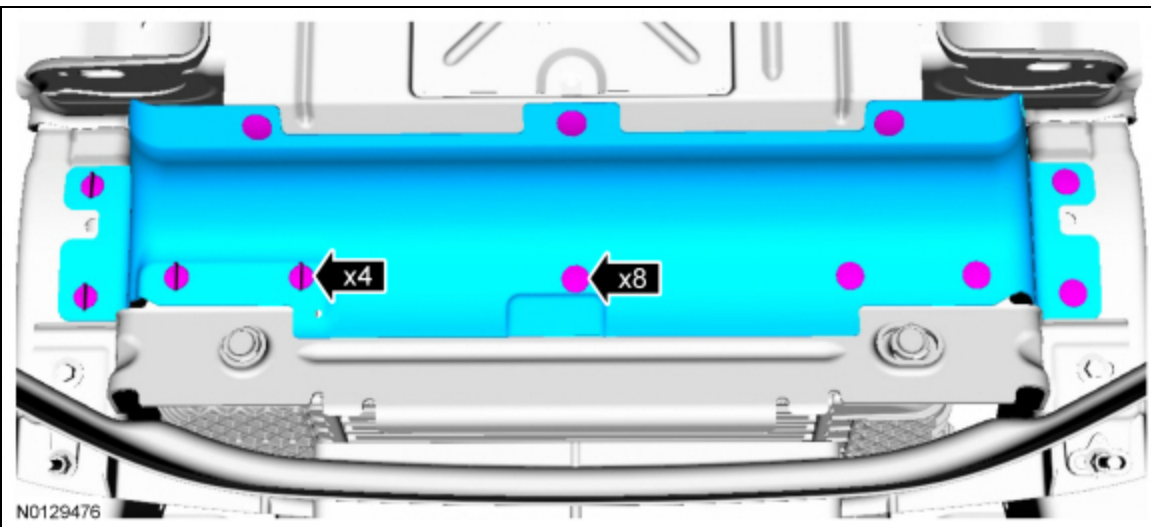
35. Install the front air dam. For additional information, refer to the Front End Body Panels — Exploded View in Section 501-02.

36. Install the skid plate and the 4 bolts.

- Tighten to 48 Nm (35 lb-ft).

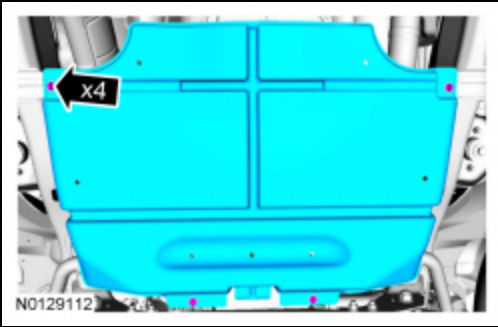
37. **NOTE:** *This step does not apply if the front engine cover is removed.*

If equipped, install the lower air deflector and the 8 pushpins. Turn the 4 quarter-turn retainers.



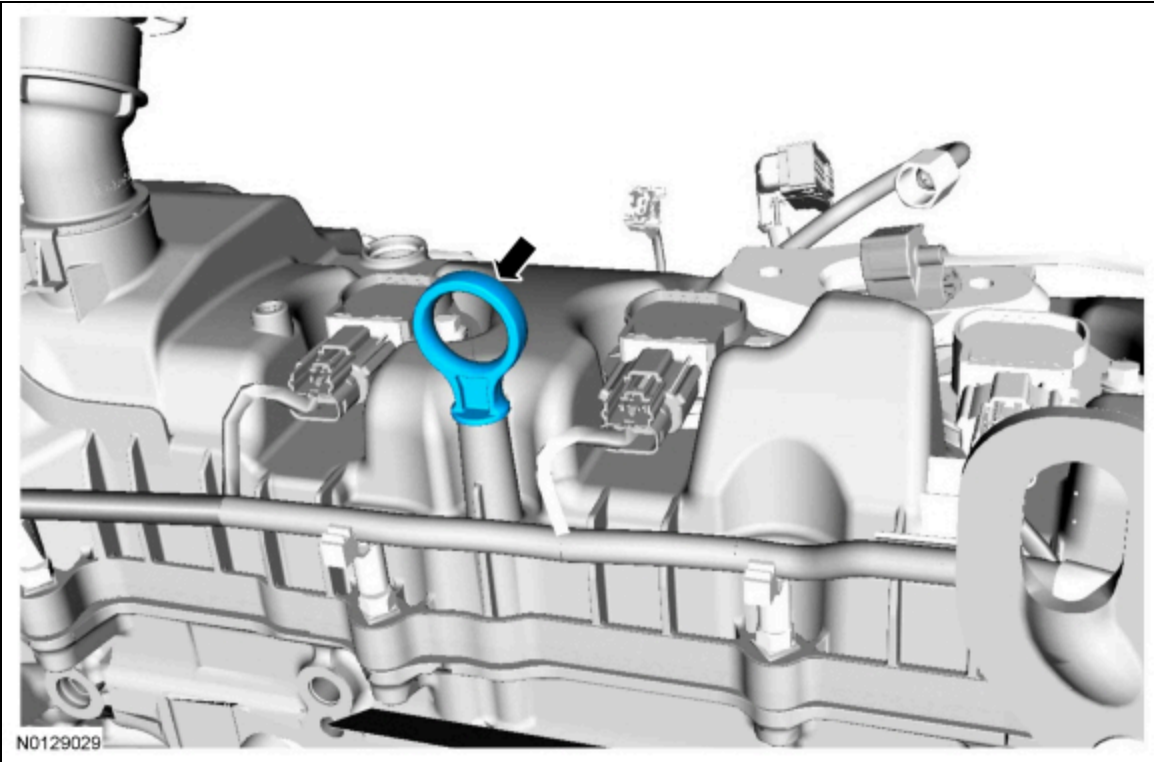
38. If equipped, install the splash shield and the 4 fasteners.





39. **NOTE:** *This step does not apply if the front engine cover is removed.*

Install the oil level indicator.



40. **NOTE:** *This step does not apply if the front engine cover is removed.*

Fill the engine with clean engine oil.

41. **NOTE:** *This step does not apply if the front engine cover is removed.*

Connect the battery ground cable. For additional information, refer to Section 414-01.